ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Submarine Control Room Watch Keeper (Dived)

Qualification Code: AIMZ

MITE CODE 123828

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CNPTG Approval – 22/04/2024
Version No: 1.5

RECORD OF AMENDMENTS

The “Authority” reference indicates approval of the corresponding chapters of this QSP document. Authority for Chapters 1 and 2 of this QSP resides with Chief of the Naval Staff (C Naval), Director Naval Training and Education (DNTE). The Curriculum Control Authority (CCA) for Chapters 3 and 4 is Canadian Forces Naval Operations School. The approval date, the date that the document is checked back in, should also appear in the bottom right corner of every page (the change being made in the footer). The Revision column is intended to provide a short indication of where the change(s) was/were made. Details on the change(s) must be amplified in Annex J “Course History.”

REVISION	APPROVAL DATE	AUTHORITY
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FOREWORD

1. This training document is developed under the authority of Commander Naval Training Group (CNTG) and in accordance with the following references:

   a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education;

   b. A-PD-055/PP-001 Occupation Specialty Specification (Officer and NCM) dated 20 October 2000.

2. This publication is effective on receipt. Performance Objectives (POs) are explained fully in Chapter 2. The Lesson Specifications (LSs)/Enabling Objectives (EOs), Enabling Checks (ECs) and Performance Checks (PCs) which specify the standards that the trainee must achieve to become qualified are listed in Chapter 4.

3. Suggestions for amendments may be initiated by anyone. Amendments must be forwarded through the normal chain of command. Specifically, training establishments are to summarize all requested amendments in writing to the applicable approving authority. QSP revision procedures are outlined in the NTI. Final approvals for amendments are as follows:

   a. Chapters 1 and 2 – Commander Naval Training Group (CNTG). Suggested amendments must first have the approval of the Curriculum Control Authority Training Establishment; and

   b. Chapters 3 and 4 – Commander Naval Training Development Center Atlantic (NTDC(A)). Training Establishments that conduct training and are not the CCA must forward suggested amendments to Naval Training Development Center.

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF NAVAP PERSONNEL TRAINING GROUP

AIM

1. The aim of this qualification standard and plan (QSP) is to provide the CommandingOfficer of a Victoria Class Submarine with a Submariner capable of performing the duties of a Control Room Watch keeper (Dived), NQual AIMZ.

2. This QSP is designed to assist instructors in preparing and organizing themselves forthe Control Room Watch Keeper (Dived) course and should be read prior to commencing any preparation or scheduling. Any suggestions for the improvement of this document should be brought to the attention of the Curriculum Control Authority at NTDC(A)

COURSE DURATION CALCULATION

3. Course duration is calculated in accordance with (IAW) NPTORD 4500-22, para 4.5. For this course, a period of instruction is 50 minutes in duration.

Course Duration Calculation		
Allocation of Actual Training Time (ATT)	Periods	Days
Lesson Guidance (LG)	162	20.25
Enabling Checks (ECs)	24	3
Performance Checks (PCs)	12	1.5
ATT SUB TOTAL:	**198**	**24.75**
Allocation of BAT, TAT, TTT and PT	Periods	Days
Base Administrative Time (BAT)	2	0.25
Training Administrative Time (TAT)	2	0.25
Training Travel Time (TTT)	16	2
Physical Fitness (PT)	16	2
BAT, TAT, TTT, PT SUB TOTAL:	**36**	**4.5**
(ATT+BAT+TTT+TAT+PT) **TOTAL:**	**234**	**29.25 days**

RESOURCES

4. Identify key resources required to conduct the training. Additional training establishment resources can be found at Annex H.

REQUIRED COURSE STAFFING				
TITLE	**RANK**	**QUALIFICATION**	**QTY**	**REMARKS**
Course Officer	Lt (N)	Victoria Class Qualified AILS	1	
Divisional Chief	CPO2	Victoria Class Qualified AILS	1	

Senior instructor	PO1/PO2	Victoria Class Qualified AILS	1	
Instructor	MS/PO2	Victoria Class Qualified AILS, AIMZ	1	

CRITICAL TRAINING RESOURCES	NO. OF DAYS
S-17 Classroom	2
S-17 SCTT	16
S-100 SCT Trainer	12

COURSE CAPACITY

5. Planning for the conduct of training is based on the following guidelines:

   a. Maximum – 8

   b. Optimum – 6

   c. Minimum – 3

COURSE PREREQUISITES

6. Candidates for this course should be from the following MOSIDS:

   a. W ENG TECH SONAR 00366-07 all ranks;

   b. W ENG TECH COMMS 00366-04 all ranks;

   c. W ENG TECH FC 00366-05 all ranks;

   d. NCI OP 00114 all ranks;

   e. NAV COMM 00299 all ranks;

   f. Physician Assistant 00374 all ranks

   g. W ENG TECH Jr 00366

   h. Sonar Op 00324 (AILE, MS and above)

RELATED DOCUMENTS

7. The following documents are related to this QSP:

   a. A-PD-055-003/PQ-001 OSS AIMZ

EVALUATION

8. Program evaluation shall be conducted in accordance with the guidance provided in A-P9-050-000 CFITES Vol. 11 Evaluation and Validation Techniques.

9. Trainee evaluation shall be conducted in accordance with NTI 03/2007

VALIDATION

10. The training will be validated every five years in accordance with Naval Training Instructions, based on the task list in Annex B.

QUALIFICATION

11. Successful completion of this course constitutes the primary means of attaining the AIMZ qualification.

PRIOR LEARNING ASSESSMENT

12. Prior Learning Assessment shall be conducted in accordance with DAOD 5031-1, Canadian Forces Military Equivalencies Program

SPECIAL INSTRUCTIONS

13. For the purposes of this QSP, Victoria Class qualification is defined as:

    a. AILS qualified

    b. Holds a minimum of 6 months operational experience.

14. Training Bookings: All classroom and Submarine Trainer bookings are booked through NFS(A) Scheduling

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF NAVAP PERSONNEL TRAINING GROUP

PO 001

1. Performance: Operate One Person Control (OPC)
2. Conditions:
   a. Given:
      (1) Operational Submarine; or
      (2) SCT.
   b. Denied:
      (1) References; and
      (2) Supervision
   c. Environmental:
      (1) At Sea;
      (2) All weather conditions;
      (3) Surfaced or submerged; and
      (4) In all light conditions.
3. Standard. The submariner shall, IAW references, operate One Person Control (OPC) (T0001), to include:
   a. Auto Mode;
   b. Manual Mode;
   c. Emergency mode during EOPs
4. References.

   A4, A5, B1.
5. Specification Task Numbers

   T0001
6. Supporting Skills

   N/A
7. Supporting Knowledge

   K0001, K0002, K0003, K0004, K0005, K0006, K0010, K0011, K0013, K0020.

8. Training Limitations

NIL

PO 002

1. Performance: Operate Submarine Fire Control System (SFCS)
2. Conditions:
   a. Given:
      (1) Operational Submarine; or
      (2) SCTT; and
      (3) Supervision.
   b. Denied:
      (1) References
   c. Environmental:
      (1) At Sea;
      (2) All weather conditions;
      (3) Surfaced or submerged; and
      (4) All light conditions.
3. Standard. The submariner shall, IAW references, operate Submarine Fire Control System (SFCS) (T0002), to include:
   a. Initializing the system;
   b. Operating FCC, to include:
      (1) Conducting threat motion analysis (TMA);
      (2) Carrying out drills and procedures; and
      (3) Performing Urgent Attack
   c. Operating Sensor Analysis Console (SAC), to include:
      (1) Conducting threat motion analysis (TMA);
      (2) Track Management.
4. References

   A1, A2, A3, B2, B3, B4
5. Specification Task Numbers

T0002

6. Supporting Skills

N/A

7. Supporting Knowledge

K0007, K0018, K0019, K0020, K0022, K0023

8. Training Limitations

NIL

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDANT / COMMANDING OFFICER OF NAVAL TRAINING DEVELOPMENT CENTER ATLANTIC

GENERAL

1. This is a pass/fail course. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and must be successfully completed to achieve this qualification.

PERFORMANCE CHECKS

2. Summative evaluation for this course is in the form of PCs conducted at the termination training for each PO. PCs are tests to assess theaccomplishment of the performance under the conditions and to the standards stated in the PO. While most of the assessments are practical, other forms of evaluation are conducted where practical assessment is not feasible or possible.

3. Conduct of any PC will be IAW NPTORD 4500-54

4. Supplemental PCs must be in the same format and have the same standard as theoriginal test. PCs may contain more than one part to facilitate different methods of assessment or to modularize particularly extensive performance requirements. Unless otherwise stipulated, failure of one part of a PC only requires retest on that part and does not require the trainee to be retested on the entire PC.

RESPONSIBILITY FOR TESTING

5. Standards cell provides a measure of quality control by confirming that POs are achieved according to the standards delineated in the QSP. They are responsible for the design, administration and scoring of all Performance Checks. Enabling Checks willbe the responsibility of the Instructional Staff.

PROGRESS MONITORING

6. Student Progress monitoring including unsatisfactory progress IAW NFS(A) SO 4063

TRAINING REVIEW BOARD

7. Training Review Board requirements are contained in NPTORD 4500-54 Training Review Board.

COURSE REPORTS

8. Course reports (CF 377) will be prepared and distributed IAW NPTORD 4500-51. Thenarrative in block 17 will indicate the trainee's strengths and weaknesses in relation to specific POs and provide information of use to the trainee's Commanding Officer and Career Manager. Training limitations or incomplete training must be identified on the course report.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDANT / COMMANDING OFFICER OF NAVAL TRAINING DEVELOPMENT CENTER ATLANTIC

EO 001.01

1. Performance: Operate the OPC in Auto Mode
2. Conditions:
   a. Given:
      (1) SCT;
      (2) References; and
      (3) Assistance/supervision as required;
   b. Denied:
      Nil;
   c. Environmental:
      (1) Victoria Class Submarine Team Trainer (VCSTT);
      (2) All simulated weather conditions;
      (3) Simulated surfaced or submerged;
      (4) All light conditions; and
      (5) All environmental conditions.
3. Standard. The trainee shall, IAW references, operate the OPC in automode, to include:
   a. Controls and indicators;
   b. Keying data;
   c. Course/depth control;
   d. Course/Depth failure;
   e. Under shipboard emergency conditions; and
   f. Voice procedures.
4. Time.
   a. 24 x 50 minute periods.
5. Method / Media.

a. 8 Lecture periods;

b. 16 Practical periods.

6. References.

c. A4, A5, B1.

7. Training Aids.

d. SCT;

e. White Board;

f. Power Point;

g. Wall Drops; and

h. Handouts.

8. Test Details.

i. EO 001.01 will be tested in EC 001.01. The material in this EO will be tested in PC001.

9. Remarks.

j. Nil.

10. Teaching Points.

TP	Description	Method	Time	Reference
TP1	Controls and Indicators to include: • Command and information panel to include: ▪ Emergency compass repeat; ▪ Heal angle indicator; ▪ Trim angle indicator; ▪ Forward hydroplane indicator; ▪ Rudder indicator; and ▪ RPM indicator. • Helm panel left toinclude: ▪ Aft plane balance angle	Explanation, Demonstration, Imitation		(A8) VCSOP (B1)OPC

	indicator; ▪ Fore plane balance angle indicator; ▪ Ratio change control; ▪ Fore planes, rudder and Aft planes, position control indicator lamps; ▪ Rudder rate emergency indicator lamp; ▪ Aft planes air emergency indicator; ▪ Dived stop control switch; ▪ Dived stop control indicator lamp; ▪ Fore planes extended indicator; ▪ Fore planes control switch; ▪ Bypass panel to include: • Bypass control switch; • Following servo/manual control switches; ▪ Aft planes; ▪ Rudder; and ▪ Fore planes. • Helm panel to include: ▪ Clock; ▪ Log readout; ▪ Echo Sounder; ▪ Pitch angle indicator; ▪ Pitch rate indicator; ▪ Depth display; ▪ Depth rate indicator; ▪ Fore and Aft plane position indicators; ▪ Fault indicators to include: • Aft planes; • Fore planes; and • Rudder. ▪ Compass repeater courseindicator; and ▪ Dimmer control. • Propulsion control panel to			

	include: ▪ Propulsion alarm indicators; ▪ Illuminated push buttons (IPBs) to include: • Main motor breakers aft and forward; • Audible off; • Execute; and • Digital test. ▪ Controlling station indicators (SCC or Propulsion switchboard: ▪ Group order switch and associated indicators; ▪ Slow speed motor generator indicator; ▪ Main motor blower indicators; • Speed order switch and associated indicators; ▪ Propulsion trimmer control; ▪ Actual RPM indicator; ▪ Ordered RPM digital indicator; and ▪ Propeller order switches. • Monitoring mode selection unit(MMSU) to include: ▪ Illuminated push buttons (IPBs) to include: ▪ Port and STBD auto reset; ▪ Port and STBD auto key/data key; ▪ Port and STBD auto course available/engaged; ▪ Port and STBD auto depth available/engaged; ▪ Port and STBD manual course reset; ▪ Port and STBD manual depth reset;			

	▪ Port and STBD manual course available/reset; ▪ Port and STBD manual depth available/reset; and ▪ Alarm Accept. • Use oil rate indicator; • Use air rate indicator; • Port and STBD auto course and depth fail split indicators; • Port and STBD manual courseand depth fail split indicators; • Power supply grade; and Maintenance panel; • Rudder emergency control to include: ▪ Emergency normal switch; and ▪ Port and STBD rudder control switch; • Set Ordered key board to include: ▪ Set display windows and selection IPBs; ▪ Specific input switches; and ▪ Ordered keypad window. • Stick wheel to include: ▪ Linear variable displacement transducers; ▪ Fore and aft plane movement; and ▪ Rudder movement.			

TP2	• Turn over report and keying data to include: • Set up for MMSU to include: ▪ Routine required to restore control in auto mode; and ▪ Recover from failure.	Explanation, Demonstration, Imitation		
TP3	• Course and Depth control to include: ▪ Take course/depth control in auto mode; and ▪ Altering course/Depth at setlimit.	Explanation, Demonstration, Imitation		(B1) OPC
TP4	• Course/depth failure to include: ▪ Recognize course/depth failureindication; and ▪ Distinguish between grade Aand grade B alarms in auto.	Explanation, Demonstration, Imitation		(A9) VCEOP, (B1) OPC
TP5	• EOPs to include: ▪ Reverting to manual modeduring emergencies.	Explanation, Demonstration, Imitation		(A9) VCEOP, (B1) OPC

TP6	• Voice procedures	Explanation, Demonstration, Imitation		(A9) VCEOP

ENABLING CHECK – EO 001.01
PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCT.

2. Oral question bank.

Arrangements and Procedures

3. EC 001.01 will be a Practical test based on content from EO 001.01. The instructor shall question the trainee on the controls and indicators of the OPC while in Auto Mode. The trainee will be required to locate, identify, and describecontrols and indicators as directed by the instructor.

4. The trainee will be required to operate the OPC in auto mode as directed by the instructor using applicable voice procedures.

5. Time required: 1 x 50 min period per student.

6. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 is a Practical test based on content from EO 001.01. The instructor will question you on the controls and indicators of the OPC while in Auto Mode. You will be required to locate, identify, and describe controls and indicators as directed by the instructor.

2. You will be required to operate the OPC in auto mode as directed by the instructor using applicable voice procedures.

3. Time required: 1 x 50 min period

4. References will not be required.

5. Pass mark IAW ECL 001.01

PART III – ENABLING CHECK NOTES

NIL

PART IV – ENABLING CHECK LIST

STUDENT NAME:_____

TYPE OF RUN: DEVELOPMENTAL:_____

DATE:_____ ASSESSMENT:_____

(use all columns) (**don't** use columns in grey)

PERFORMANCE FACTORS	SAT	UNSAT	Comments
Critical Performance factors			
1. Telegraph orders and indicators			
Non-Critical Performance Factors			
1. Course control;			
2. Depth control;			
3. Controls and indicators:			
a. Command and information panel			
b. Helm panel left			
c. Helm panel			
d. Monitoring mode selection unit			
e. Propulsion panel			
4. Under shipboard emergency conditions			
5. Voice procedures			
6. Course/depth control failure			

ASSESSORS / TACTICS REMARKS:

OVERALL ASSESSMENT:

Are the critical performances passed?	
YES	**NO**
Are 7 of 10 non-critical performance factors satisfactory?	
YES	**NO**
PASS	**FAIL**

STUDENTS SIGNATURE **STAFF ASSESSOR**

THIS FORM TO BE RETAINED ON COURSE FILES IAW NFS(A)SSO'S

Unsatisfactory (U)	**SATISFACTORY (S)**
- No/Limited understanding of mechanics/requirements of task - Performed task but required assistance to avoid critical errors	- Understood mechanics of the task - No critical errors made - Performed task with minimalassistance/coaching

EO 001.02

1. Performance: Operate the OPC in Manual Mode

2. Conditions:

   a. Given:

      (1) SCT;

      (2) References; and

      (3) Assistance/supervision as required;

   b. Denied:

      (1) Nil;

   c. Environmental:

      (1) VCSTT;

      (2) All simulated weather conditions;

      (3) Simulated surfaced or submerged;

      (4) All light conditions; and

      (5) All environmental conditions.

3. Standard. The trainee shall, IAW references, operate the OPC in manualmode, to include:

   a. Course/depth control;

   b. Course/depth failure; and

   c. Under ship board emergency conditions.

4. Time.

   a. 20 x 50 minute periods.

5. Method / Media.

   a. 4 Lecture periods;

   b. 16 Practical periods.

6. References.

   a. A4, A5, B1.

7. Training Aids.

   a. SCT.

8. Test Details.

   a. EO 001.02 will be tested in EC 001.02. The material taught in this EO will be tested in PC 001.

9. Remarks.

   a. Nil.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	Course and depth control ▪ Plane in manual mode ▪ Propulsion system ▪ Helm panel center interpretation ▪ Dive stop control ▪ Fore planes control ▪ Bypass panel ▪ Ratio Control Planing (CP Mode)	Explanation, Demonstration, Imitation		(A8) VCSOP
TP2	Course and depth failure ▪ Recognizing ▪ Reacting	Explanation, Demonstration, Imitation		(A9) VCEOP, (B1) OPC
TP3	Under ship board emergency conditions	Explanation, Demonstration, Imitation		(A9) VCEOP, (B1) OPC

ENABLING CHECK – EO 001.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCT.

2. Scenario.

Arrangements and Procedures

1. EC 001.02 will be a Practical test based on content from EO 001.02. The trainee will be required to operate the OPC in manual mode as directed by the instructor using applicable voice procedures.

2. Pass mark is IAW ECL 001.02.

3. Time required: 1 x 50 min period per student.

4. References will not be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.02 will be a Practical test based on content from EO 001.02. You will be required to operate the OPC in manual mode as directed by the instructor using applicable voice procedures.

2. Pass mark is IAW ECL 001.02.

3. Time required: 1 x 50 min period.

4. References will not be required.

PART III – ENABLING CHECK NOTES

5. NIL

PART IV – ENABLING CHECK LIST

STUDENT NAME:_____

TYPE OF RUN: DEVELOPMENTAL:_____

DATE:_____ ASSESSMENT:_____

(use all columns) (**don't** use columns in grey)

PERFORMANCE FACTORS	**SAT**	**UNSAT**	**COMMENTS**
Critical Performance factors			
1. Telegraph orders and indicators			
Non-Critical Performance Factors			
1. Course control;			
5. Depth control;			
6. Voice procedures			
7. Course/depth control failure			

ASSESSORS / TACTICS REMARKS:

OVERALL ASSESSMENT:

Are the critical performances passed?	
YES	**NO**
Is 1 of the 4 non-critical performance factors satisfactory?	
YES	**NO**
PASS	**FAIL**

STUDENTS SIGNATURE **STAFF ASSESSOR**

THIS FORM TO BE RETAINED ON COURSE FILES IAW NFS(A)SSO'S

Unsatisfactory (U)	**SATISFACTORY (S)**
-No/Limited understanding of mechanics/requirements of task - Performed task but required assistance to avoid critical errors	- Understood mechanics of the task - No critical errors made - Performed task with minimalassistance/coaching

EO 001.03

1. Performance: Operate the OPC during EOPs.
2. Conditions:
   a. Given:
      (1) SCT;
      (2) References; and
      (3) Assistance/supervision as required;
   b. Denied:
      (1) Nil;
   c. Environmental:
      (1) VCSTT;
      (2) All simulated weather conditions;
      (3) Simulated surfaced or submerged;
      (4) All light conditions; and
      (5) All environmental conditions.
3. Standard. The trainee shall, IAW references, operate the OPC during EOPs.
4. Time.
   a. 18 x 50 minutes periods.
5. Method / Media.
   a. 2 Lecture periods;
   b. 16 Practical periods.
6. References.
   a. A5.
7. Training Aids.
   a. SCT;
   b. Whiteboard;

c. PowerPoint;

d. Wall drops; and

e. Handouts.

8. Test Details.

a. This EO will be tested in EC 001.03. The material in this EO will be tested in PC 001.

9. Remarks.

a. Nil.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	• EOPs to include: ▪ Planes jams; ▪ Emergency go deep; ▪ Hydraulic failure/burst; ▪ Flood ▪ Fire; ▪ HP air burst; and ▪ Collision/grounding. ▪ Electrical Failures.	Explanation, Demonstration, Imitation	18 x 50 Minute periods	(A9) VCEOP

ENABLING CHECK – EO 001.03

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCT.
2. Scenario.

Arrangements and Procedures

1. EC 001.03 will be a Practical test based on content from EO 001.03.
2. Pass mark will be IAW ECL 001.03.
3. Time required: 1 x 50 min period per student.
4. EOPs will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.03 will be a Practical test based on content from EO 001.03.
2. If you have any questions during the exam, raise your hand.
3. EOPs will be provided.
4. Pass mark is IAW ECL 003.01.
5. You will have 1 x 50 min period to complete this exam.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

NIL.

PART IV – ENABLING CHECK LIST

STUDENT NAME:_____

TYPE OF RUN: DEVELOPMENTAL:_____

DATE:_____ ASSESSMENT:_____

(use all columns) (**don't** use columns in grey)

PERFORMANCE FACTORS	**SAT**	**UNSAT**	**COMMENTS**
Critical Performance factors			
1. Planes jam			
2. Emergency go deep			
3. Flood			
4. Fire			
5. Electrical failure			
6. Hydraulic burst			
7. Telegraph and order indicators			
Non-Critical Performance Factors			
1. Course control;			
2. Depth control;			
3. Voice procedures.			

ASSESSORS / TACTICS REMARKS:

OVERALL ASSESSMENT:

Are the critical performances passed?	
YES	**NO**
Are 2 of 3 non-critical performance factors satisfactory or above?	
YES	**NO**
PASS	**FAIL**

STUDENTS SIGNATURE **STAFF ASSESSOR**

THIS FORM TO BE RETAINED ON COURSE FILES IAW NFS(A)SSO'S

Unsatisfactory (U)	**SATISFACTORY (S)**
-No/Limited understanding of mechanics/requirements of task - Performed task but required assistance to avoid critical errors	- Understood mechanics of the task - No critical errors made - Performed task with minimalassistance/coaching

PERFORMANCE CHECK – PO 001

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCT.
2. Scenario.

Arrangements and Procedures

1. PC 001 will be a practical test based on content from PO 001.
2. Pass mark will be IAW PCL 001.
3. Time required: 1 x 50 min period per student**.**
4. EOPs will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001 will be a Practical test based on content from PO 001.
2. If you have any questions during the exam, raise your hand.
3. EOPs will be provided.
4. Pass mark is IAW PCL 001.
5. You will have 1 x 50 min period to complete this exam.
6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

NIL

PART IV – PERFORMANCE CHECK LIST

STUDENT NAME:_____

TYPE OF RUN: ASSESSED:_____

DATE:_____ ASSESSMENT:_____

(use all columns) (**don't** use columns in grey)

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
Critical Performance factors			
1. Telegraph orders and indicators			
2. EOPs			
Non-Critical Performance Factors			
1. Voice procedure			
2. Dive the boat achieve PD			
3. Depth +/- 1 Meter			
4. Operate OPC in AUTO			
5. Operate OPC in MANUAL			
6. Course +/- 5 Deg			
7. In MANUAL go deep			
8. Return to PD			
9. Surface the boat			

ASSESSORS / TACTICS REMARKS:

OVERALL ASSESSMENT:

Are the critical performances passed?	
YES	**NO**
Are 7 of 9 non-critical performance factors satisfactory?	
YES	**NO**
PASS	**FAIL**

STUDENTS SIGNATURE **STAFF ASSESSOR**

THIS FORM TO BE RETAINED ON COURSE FILES IAW NFS(A)SSO'S

Unsatisfactory (U)	**SATISFACTORY (S)**
-No/Limited understanding of mechanics/requirements of task - Performed task but required assistance to avoid critical errors	- Understood mechanics of the task - No critical errors made - Performed task with minimalassistance/coaching

EO 002.01

1. Performance: Introduction to Broad Band Target Motion Analysis (TMA)
2. Conditions:
   a. Given:
      (1) Supervision;
      (2) References;
   b. Denied:
      (1) Nil;
   c. Environmental:
      (1) Classroom.
3. Standard. The trainee shall, IAW references, Explain the process of Broad Band TMA to include:
   a. Line Of Sound;
   b. 1936 Ranging;
   c. How Target Solutions Are Calculated; and
   d. Time Bearing Plots.
4. Time.
   a. 4 x 50 minute periods.
5. Method / Media.
   a. 4 Lecture periods;
6. References.
   a. A1
7. Training Aids.
   a. Whiteboard;
   b. PowerPoint;
   c. Wall drops; and

d. Handouts.

8. Test Details.

a. This EO will not be tested; knowledge is demonstrated in PO 002.

9. Remarks.

a. Nil.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	• Line of Sound	Interactive lecture		(A5) NCPM 401
TP2	• 1936 Ranging ▪ Single leg eklund; ▪ Double leg eklund;	Interactive lecture		(A5) NCPM 401

TP3	• Calculating target solutionsto include: ▪ OSA; ▪ TSA; ▪ RSA; ▪ Bearing rate; ▪ Range; DOT; ▪ Speed; ▪ ATB; and ▪ Rel. Bearing.	Interactive lecture		(A5) NCPM 401
TP4	• Time Bearing Plots ▪ Opening and Closing Curves ▪ Leading and Lagging Solutions	Interactive lecture		(A5) NCPM 401

EO 002.02

1. Performance: Introduction to Narrow Band TMA
2. Conditions:
   a. Given:
      (1) Supervision; and
      (2) References;
   b. Denied:
      (1) Nil;
   c. Environmental:
      (1) Classroom.
3. Standard. The trainee shall, IAW references, explain the process of Narrow Band TMA to include:
   a. Line Of Sound WRT Frequency;
   b. Doppler;
   c. 1959 Ranging;
   d. Frequency relationships;
   e. How target solutions are calculated; and
   f. Time Frequency Plots.
4. Time.

   6 x 50 minute periods.
5. Method / Media.
   a. 6 periods Lecture;
6. References.
   a. A1.
7. Training Aids.
   a. Whiteboard;

b. PowerPoint;

c. Wall drops; and

d. Handouts.

8. Test Details.

This EO will not be tested; knowledge is demonstrated in PO 002.

9. Remarks.

a. Nil.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	• Line of Sound	Interactive lecture		(A5) NCPM 401
TP2	• What is Doppler	Interactive lecture		(A5) NCPM 401
TP3	• Frequency Relationships ▪ Fo; ▪ Fr; and ▪ Fc.	Interactive lecture		(A5) NCPM 401
TP4	• 1959 Ranging	Interactive lecture		(A5) NCPM 401
TP5	• How target solutions are calculated ▪ OSI; ▪ TSI; and ▪ RSI.	Interactive lecture		(A5) NCPM 401
TP6	• Time Frequency Plots ▪ Shift up ▪ Shift down ▪ Resolving for Ambiguity	Interactive lecture		(A5) NCPM 401

EO 002.03

1. Performance: Initialize the SFCS system.
2. Conditions:
   a. Given:
      (1) Operational SFCS;
      (2) Supervision; and
      (3) References;
   b. Denied:
      (1) Nil;
   c. Environmental:
      (1) VCSTT;
      (2) All simulated weather conditions;
      (3) Simulated surfaced or submerged; and
      (4) All light conditions.
3. Standard. The trainee shall, IAW references, initialize the SFCS to include:
   a. Normal system line up;
   b. Boot up the system; and
   c. Page setups.
4. Time.
   a. 12 x 50 minute periods.
5. Method / Media.
   a. 4 Lecture periods;
   b. 8 Practical periods.
6. References.
   a. A2.
7. Training Aids.

a. Operational SFCS;

b. Whiteboard;

c. PowerPoint;

d. Wall drops; and

e. Handouts.

8. Test Details.

a. The material in this EO will not be tested, it is an implied task in the performance being assessed in PC 002.

9. Remarks.

a. Nil.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	• Normal system line up to include: ▪ System power line up; ▪ FCC mode switches selection; and ▪ Power up.	Explanation, Demonstration, Imitation		(A6) Submarine Fire Control System IETM for all TPs
TP2	• Page set up to include: ▪ ALR and DGS recording.	Explanation, Demonstration, Imitation		
TP3	• Page Configuration Intro	Explanation, Demonstration, Imitation		

EO 002.04

1. Performance: Operate Fire Control Console (FCC) for Broad Band (BB) TMA

2. Conditions:

   a. Given:

      (1) Operational SFCS;

      (2) Supervision; and

      (3) References;

   b. Denied:

      (1) Nil;

   c. Environmental:

      (1) All simulated weather conditions;

      (2) Simulated surfaced or submerged; and

      (3) All light conditions.

3. Standard. The trainee shall, IAW references, operate FCC for Broad Band (BB) TMA to include:

   a. Conducting Target Motion Analysis (TMA);

   b. Carry out drills and procedures to include:

      (i) RMS;

      (ii) Ranging maneuver;

      (iii) Contact ZIG procedure;

      (iv) Target set up;

      (v) Range for Me;

      (vi) Returning to periscope depth;

      (vii) Close quarters.

   c. Performing Urgent Attack.

4. Time.

a. 34 x 50 minute periods.

5. Method / Media.

a. 2 Lecture periods;

b. 32 Practical periods.

6. References.

a. A2, A3, B2, B3, B4.

7. Training Aids.

a. FCC;

b. Whiteboard;

c. PowerPoint;

d. Wall drops; and

e. Handouts.

8. Test Details.

a. This EO will be tested in EC 002.04. The material in this EO will be tested in PC 002.

9. Remarks.

a. Nil.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	• Perform TMA to include: ▪ Describe the TMA pages; ▪ Perform BB TMA; ▪ Using Legs ▪ Geo Mate ▪ Min/Max Crossover Range ▪ P Boot Page	Explanation, Demonstration, Imitation		(A5) NCPM 401 Ch2 (A7) CANSADS Ch3 pg 3-2

	▪ Leading/Lagging solutions ▪ Endpoint Employment ▪ Situational Awareness ▪ Shipping Lanes ▪ Track Classifications ▪ Sensor Capabilities			
TP2	• Conduct drills and procedures to include: ▪ RMS; ▪ Ranging manoeuvre. ▪ ZIG procedure; ▪ Target Set up; ▪ Range For Me ▪ Return to PD; and ▪ Close quarters.	Explanation, Demonstration, Imitation		(A7) CANSADS Ch3 pg 3-2, 3-3, 4-1
TP3	• Perform Urgent Attack ▪ Tube Selection ▪ Threat Assignment ▪ Tube Operation ▪ Voice Procedures	Explanation, Demonstration, Imitation		A6, A7, B7, B8, B9

ENABLING CHECK – EO 002.04
PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SFCS.
2. Scenario.
3. Trainer.

Arrangements and Procedures

1. EC 002.04 will be a practical test based on content from EO 002.04.
2. Pass mark for EC is IAW ECL 002.04.
3. Time required: 1 x 50 min period per 3 students.
4. References will not be required.
5. Training aids will be required.
6. One student per console, one assigned contact per student.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.04 will be a practical test based on content from EO 002.04.
2. If you have any questions during the exam, raise your hand.
3. References will not be provided.
4. Training aids will be provided.
5. Pass mark for this EC is IAW ECL 002.04.
6. You will have 1 x 50 min period to complete this exam.
7. You will each be assigned a different contact.
8. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

NIL

PART IV – ENABLING CHECK LIST

STUDENT NAME:_____

TYPE OF RUN: DEVELOPMENT:____

DATE:_____ ASSESSMENT:_____

(use all columns) (**don't** use columns in grey)

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
Performance factors			
1. Describe the TMA pages			
2. Perform TMA to include:			
a. RMS			
b. P boot			
c. Ranging maneuver			
3. Conduct drills and procedures to include			
a. Initial page management			
b. RMS			
c. Ranging maneuver			
d. Zig procedure			
e. Visual setup			
f. Return to PD			
g. Close quarters			
4. Perform Urgent Attack			

ASSESSORS / TACTICS REMARKS:

OVERALL ASSESSMENT:

Are 9 of 12 performance factors satisfactory?	
YES	**NO**
PASS	**FAIL**

STUDENTS SIGNATURE **STAFF ASSESSOR**

THIS FORM TO BE RETAINED ON COURSE FILES IAW NFS(A)SSO'S

Unsatisfactory (U)	**SATISFACTORY (S)**
-No/Limited understanding of mechanics/requirements of task - Performed task but required assistance to avoid critical errors	- Understood mechanics of the task - No critical errors made - Performed task with minimalassistance/coaching

EO 002.05

1. Performance: Operate Fire Control Console (FCC) for Narrow Band (NB) TMA.
2. Conditions:
   a. Given:
      (1) Operational SFCS;
      (2) Supervision; and
      (3) References;
   b. Denied:
      (1) Nil;
   c. Environmental:
      (1) VCSTT;
      (2) All simulated weather conditions;
      (3) Simulated surfaced or submerged; and
      (4) All light conditions.
3. Standard. The trainee shall, IAW references, operate FCC for Narrow Band (NB) TMA to include:
   a. Conducting Target Motion Analysis (TMA);
   b. Carry out drills and procedures to include:
      (1) RMS;
      (2) Ranging manoeuvre;
      (3) Contact ZIG procedure; and
      (4) Returning to periscope depth.
4. Time.
   a. 34 x 50 minute periods.
5. Method / Media.
   a. 2 Lecture periods; and

b. 32 Practical periods.

6. References.

a. A2, A3, B2, B3, B4.

7. Training Aids.

a. FCC;

b. Whiteboard;

c. PowerPoint;

d. Wall drops; and

e. Handouts.

8. Test Details.

a. This EO will be tested in EC 002.05. The material in this EO will be tested in PC 002.

9. Remarks.

a. Nil.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	• Resolve for ambiguity ▪ True bearing ▪ Refracted bearing ▪ Ranging Maneuver	Explanation, Demonstration, Imitation		(A5) NCPM 401 (A7) CANSADS
TP2	• Frequency Shifts ▪ Shift up ▪ Shift down ▪ Zig procedure	Explanation, Demonstration, Imitation		(A5) NCPM 401 (A7) CANSADS

TP3	• Situational Awareness ▪ Steady up time ▪ Returning to PD	Explanation, Demonstration, Imitation		(A5) NCPM 401 (A7) CANSADS

ENABLING CHECK – EO 002.05
PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SFCS.
2. Scenario.
3. SCTT.

Arrangements and Procedures

1. EC 002.05 will be a practical test based on content from EO 002.05.
2. Pass mark will be IAW ECL 002.05.
3. Time required: 1 x 50 min period per 3 students.
4. References will not be required.
5. Training aids will be required.
6. One student per console, one assigned contact per student.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.05 will be a practical test based on content from EO 002.05.
2. References will not be provided.
3. Training aids will be provided.
4. Pass mark is IAW ECL 002.05.
5. You will have 1 x 50 min period to complete this exam.
6. You will be each assigned a different contact.
7. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

NIL

PART IV – ENABLING CHECK LIST

STUDENT NAME:_____

TYPE OF RUN: DEVELOPMENTAL:_____

DATE:_____ ASSESSMENT:_____

(use all columns) (**don't** use columns in grey)

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
Performance factors			
1. Describe the TMA pages			
2. Perform TMA to include:			
a. RMS			
b. P boot			
c. Ranging manoeuvre			
d. Opening/Closing solution			
3. Conduct drills and procedures to include:			
a. Initial page management			
b. RMS			
c. Ranging manoeuvre			
d. Zig procedure			
e. Resolve for Ambiguity			
f. Return to PD			

ASSESSORS / TACTICS REMARKS:

OVERALL ASSESSMENT:

Are 8 of 11 performance factors satisfactory?	
YES	**NO**
PASS	**FAIL**

STUDENTS SIGNATURE **STAFF ASSESSOR**

THIS FORM TO BE RETAINED ON COURSE FILES IAW NFS(A) SSO'S

Unsatisfactory (U)	**SATISFACTORY (S)**
-No/Limited understanding of mechanics/requirements of task - Performed task but required assistance to avoid critical errors	- Understood mechanics of the task - No critical errors made - Performed task with minimalassistance/coaching

EO 002.06

1. Performance: Operate Contact Development Coordinator (CDC) Console.
2. Conditions:
    a. Given:
        (1) Operational CDC Console;
        (2) Supervision; and
        (3) References;
    b. Denied:
        (1) Nil;
    c. Environmental:
        (1) All simulated weather conditions;
        (2) Simulated surfaced or submerged; and
        (3) All light conditions.

3. Standard. The trainee shall, IAW references, Operate Contact Development Coordinator Console to include:
    a. Conducting Threat Motion Analysis (TMA);
    b. Target set ups; and
    c. Track management.
4. Time.
    a. 10 x 50 minute periods.
5. Method / Media.
    a. 2 Lecture periods;
    b. 8 Practical periods.
6. References.
    a. A1, A3.
7. Training Aids.

a. CDC Console;

b. Whiteboard;

c. PowerPoint;

d. Wall drops; and

e. Handouts.

8. Test Details.

a. This EO will be tested in EC 002.06. The material in this EO will be tested in PC 002.

9. Remarks.

a. Nil.

10. Teaching Points.

	Description	Method	Time	Reference
TP1	• Sensor Correlation ▪ AIS ▪ Periscope ▪ ESM ▪ Radar ▪ PRS ▪ Bow ▪ AI&R ▪ Flank ▪ Towed Array	Explanation, Demonstration, Imitation		(A5) NCPM 401 (A7) CANSADS
TP2	• Information Management ▪ Situational Awareness	Explanation, Demonstration, Imitation		(A5) NCPM 401 (A7) CANSADS

TP3	• Perform track managementto include: ▪ Threat nomination ▪ Track designation ▪ Target setup ▪ Surface sitrep ▪ Warner reports ▪ CDC annotation.	Explanation, Demonstration, Imitation		(A5) NCPM 401 (A7) CANSADS

ENABLING CHECK – EO 002.06

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. CDC Console.
2. Scenario.
3. SCTT.

Arrangements and Procedures

1. EC 002.06 will be a practical test based on content from EO 002.06.
2. Pass mark will be IAW ECL 002.06.
3. Time required: 1 x 50 min period per 3 students.
4. References will not be required.
5. One student per console, one assigned contact per student.
6. Training aids will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.06 will be a practical test based on content from EO 002.06.
2. References will not be provided.
3. Training aids will be provided.
4. Pass mark is IAW ECL 002.06.
5. You will have 1 x 50 min period to complete this exam.
6. You will each be assigned a different contact.
7. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

NIL

PART IV – ENABLING CHECK LIST

STUDENT NAME:_____

TYPE OF RUN: DEVELOPMENTAL:_____

DATE:_____ ASSESSMENT:_____

(use all columns) (**don't** use columns in grey)

PERFORMANCE FACTORS	SAT	UNSAT	COMMENTS
Performance factors			
1. Perform Sensor Correlation to include:			
a. AIS			
b. Periscope			
c. ESM			
d. Radar			
e. Sonar			
2. Perform Return to PD procedures			
3. Perform track management to include:			
a. Threat nomination			
b. Track designation			
c. Voice procedures			
d. CDC annotation			

ASSESSORS / TACTICS REMARKS:

OVERALL ASSESSMENT:

Are 8 of 10 performance factors satisfactory?	
YES	**No**
PASS	**FAIL**

STUDENTS SIGNATURE **STAFF ASSESSOR**

THIS FORM TO BE RETAINED ON COURSE FILES IAW NFS(A)SSO'S

Unsatisfactory (U)	**SATISFACTORY (S)**
-No/Limited understanding of mechanics/requirements of task - Performed task but required assistance to avoid critical errors	- Understood mechanics of the task - No critical errors made - Performed task with minimalassistance/coaching

PERFORMANCE CHECK – PO 002

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SFCS.
2. Scenario.
3. SCTT.

Arrangements and Procedures

1. PC 002 will be a practical test based on content from PO 002.
2. Pass mark will be IAW PCL 002.
3. Time required: 1 x 50 min period per student.
4. References will not be required.
5. Training aids will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 002 will be a practical test based on content from PO 002.
2. References will not be provided.
3. Training aids will be provided.
4. Pass mark is IAW PCL 002.
5. You will have 1 x 50 min period to complete this exam.
6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

NIL

PART IV – ENABLING CHECK LIST

STUDENT NAME:_____

TYPE OF RUN: DEVELOPMENTAL:_____

DATE:_____ ASSESSMENT:_____

(use all columns) (**don't** use columns in grey)

PERFORMANCE FACTORS	**Sat**	**Unsat**	**COMMENTS**
Critical Performance factors			
1. Perform Target Set Up			
2.Operate TMA To Include			
a. Ranging Maneuver +- 10% Accuracy.			
b. Zig Procedure +- 15% Accuracy			
. c. Close Quarters Drill			
Non-Critical Performance Factors			
1. Operate TMA to include:			
a. RMS			
b. P boot			
2. Perform Urgent Attack			
3. Perform track management to include:			
a. Threat nomination			
b. Track designation			
c. Voice procedures			
d. CDC annotation			

ASSESSORS / TACTICS REMARKS:

OVERALL ASSESSMENT:

Are the critical performances passed?	
YES	**NO**
Are 4 of 7 non-critical performance factors satisfactory	
YES	**No**
PASS	**FAIL**

STUDENTS SIGNATURE **STAFF ASSESSOR**

THIS FORM TO BE RETAINED ON COURSE FILES IAW NFS(A)SSO'S

Unsatisfactory (U)	**SATISFACTORY (S)**
-No/Limited understanding of mechanics/requirements of task - Performed task but required assistance to avoid critical errors	- Understood mechanics of the task - No critical errors made - Performed task with minimal assistance/coaching.

ANNEX A – MAIN REFERENCES

A = CF or other official Canadian government publication, or

Second and subsequent references in any category

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other publications

Reference Code	Document Number	Document Title
A1	NCPM 401	Naval Combat Procedures Manual
A2	C-69-815-000/MS-001	Submarine Fire Control System IETM
A3	CANSADS	Canadian Submarine Attack Drills
A4	VCSOP	Victoria Class Standard Operating Procedures
A5	VCEOP	Victoria Class Emergency Operating Procedures
B1	BRF 1966(21) 02	One Person Control Console (OPC)
B2	C-69-699-000/MS-001	Submarine Fire Control System Manual Vol.1
B3	C-69-699-000/MS-002	Submarine Fire Control System Manual Vol. 2
B4	C-69-699-000/MS-003	Submarine Fire Control System Manual Vol. 3

ANNEX B – TASK LIST

JBOS/OSS TASK NO.	TASK ELEMENT	TRG / NO TRG	PO
T0001	Operate One Person Control (OPC)	Train	001
T0002	Operate Submarine Fire Control System (SFCS)	Train	002
T0003	Operate Manual Plots	No Train	

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates:

Location:

Board Members:

SURNAME	INITIALS	RANK	MOSID	UNIT

Board Objective:

Specification Task – No Train Decisions:

1. The tasks selected for training are detailed at Annex B. Tasks not selected for training are listed below with substantiation:

TASK SERIAL	TASK ELEMENT	SUBSTANTIATION

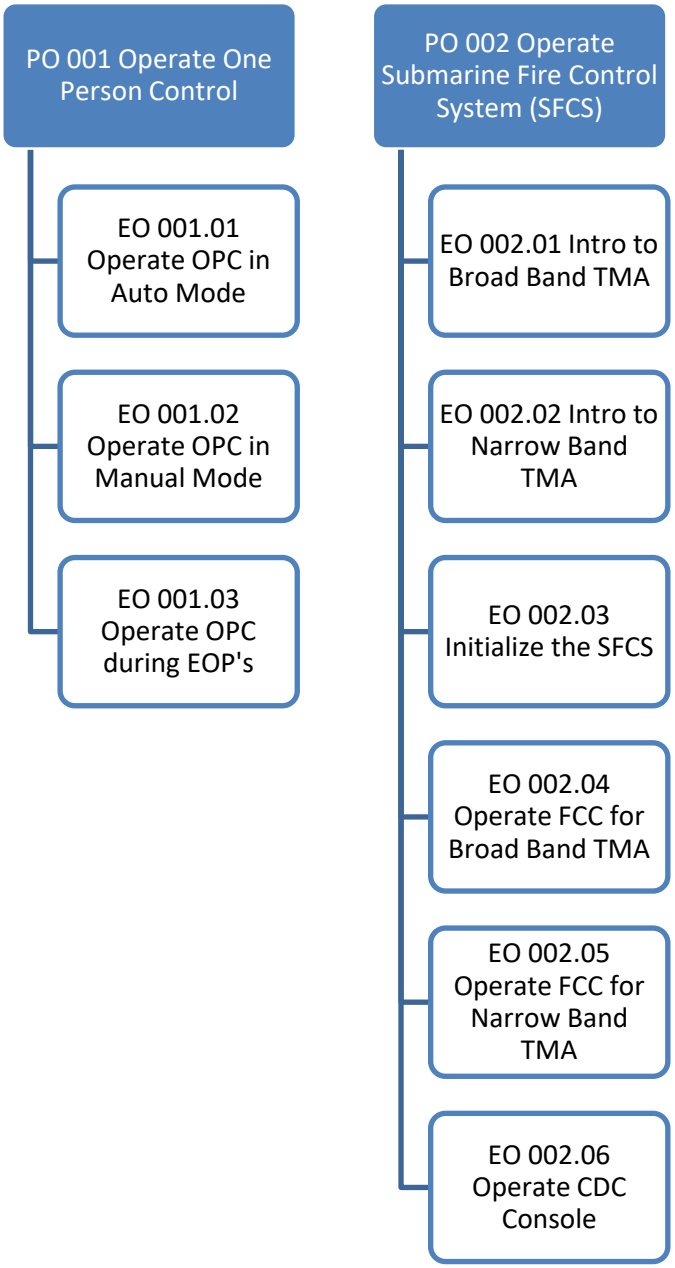
Discussions / Decisions:

Lessons Learned:

Guidance to Instructional Designers:
Follow-up Action:

Other Considerations (if applicable)

ANNEX D – QUALIFICATION STANDARD AND PLAN WRITING BOARD SCALAR



ANNEX E – ON-JOB PERFORMANCE RECORD

There is no OJT for this qualification

ANNEX F – ACRONYMS / ABBREVIATIONS

ACRONYM / ABBREVIATION	TERM/DEFINITION
BFP	BEARING FREQUENCY PLOT
CDC	CONTACT DEVELOPMENT COORDINATOR
CEP	CONTACT EVALUATION PLOT
EOP	EMERGENCY OPERATING PROCEDURES
FCC	FIRE CONTROL CONSOLE
OPC	ONE PERSON CONSOLE
SAC	SENSOR ANALYSIS CONSOLE
SCT	SHIP CONTROL TRAINER
SCTT	SUBMARINE COMMAND TEAM TRAINER
SFCS	SUBMARINE FIRE CONTROL SYSTEM
SSE	SUBMERGED SIGNAL EJECTOR
TBP	TIME BEARING PLOT
TMA	THREAT MOTION ANALYSIS
VCSTT	VICOTRIA CLASS SUBMARINE TEAM TRAINER
WAC	WEAPONS ANALYSIS CONSOLE
WSC	WEAPONS STOWAGE COMPARTMENT

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

PO/EO	OBJECTIVE	EO		EC	PC	PO
		L	P			
PO 001	**Operate One Person Control (OPC)**	**14**	**48**	**18**	**6**	**86**
EO 001.1	Operate the OPC in AUTO Mode	8	16	6		
EO 001.2	Operate the OPC in MANUAL Mode	4	16	6		
EO 001.3	Operate the OPC during EOPs	2	16	6	6	
PO 002	**Operate Submarine Fire Control System (SFCS)**	**20**	**80**	**6**	**6**	**112**
EO 002.1	Intro To BB TMA	4	0			
EO 002.2	Intro To NB TMA	6	0			
EO002.3	Initialize the SFCS system	4	8			
EO 002.4	Operate Fire Control Console (FCC) for BB TMA	2	32	2		
EO 002.5	Operate Fire Control Console (FCC) NB TMA	2	32	2		
EO 002.6	Operate Sensor Analysis Console (SAC)	2	8	2	6	
TOTAL		**34**	**128**	**24**	**12**	**198**

Note

- Sequencing and Timings table shown above is calculated for an optimum course load of 6 students.
- For the CRWK(D) course, since all exams are practical and assessed individually, each student has 1 period to complete each ECs and PCs.
- Course schedule sequencing and timings will change based on the number of students attending the CRWK(D) course.

ANNEX H – TRAINING RESOURCES

In accordance with Chapter 1 paragraph 3 add any extra tables or paragraphs used for Resources in this annex. (**Not** to be deleted.)

ANNEX I – COSTING INFORMATION

This table provides an estimated cost to deliver this training. Considered costs include Temporary Duty (TD), material, and personnel costs. TD costs include Rations & Quarters (R&Q) costs for relevant TE, and National Joint Counsel factors (meals & incidentals). Material costs are best estimates.

TD Costs	# Trainees + Instructors /	Unit Price	# Days	Initial Course	Each Successive Course	5 Year Total
Airfare (Return)	6	$1,000.00	n/a	$6,000.00	$6000.00	$30,000.00
R&Q	6	$72.31	30	$13,015.80	$13,015.80	$65,079.00
Incidentals	6	$17.50	30	$3,150.00	$3,150.00	$15,750.00
Subtotal				**$22,165.80**	**$22,165.80**	**$110,829.00**
Material Costs	**Quantity**	**Unit Price**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
PP&S (Estimate)	16	$50.00	1	$800	$800	$4,000.00
Instructional Materials (Printed / Handouts etc.)	15	$30.00	1	$450	$450	$2,250.00
Subtotal				**$1,250.00**	**$1,250.00**	**$6,250.00**
Personnel Costs	**Quantity**	**Cost/day**	**# Days**	**Initial Course**	**Each Successive Course**	**5 Year Total**
N/A						
Subtotal						
Total Estimated Costs				**$23,415.80**	**$23,415.80**	**$117,079.00**

ANNEX J – COURSE HISTORY

(April 2008) This course was amended to remove PO 003 Operate Submarine Radar as it is no longer a responsibility/task of the Control Room Watch Keeper (Dived). An SPQR amendment and QSP Change Requests have been submitted to ensure alignment between documentation and that proper nomenclature is reflected.

(April 2008) Submarine Fire Control System nomenclature was amended throughout this document to bring it in line with recent system updates.

Further to this several the sequencing and timings have been amended to reflect actual instructional time as it occurs in the current course. This course was never submitted to a Pilot Course Evaluation Process and over the years updates were never carried out to the document. That said, the findings of a recent Programme Evaluation conducted in CFNOS (Jan – Mar 08) concluded that the course as delivered is providing the desired training to candidates. Given this fact, it was determined by CFNOS, with concurrence from DMTE, to reverse engineer the required training updates from the training to the QSP and Specialty Specification. This was done to ensure that all documentation was aligned.

(Nov 2008) version 1.02 involved minor typographical error correction done in EO 002.01

(Dec 2009) Version 1.3 was approved to remove PO 004 – Perform the duties of Forward Watch Keeper, from the CRWK(D) QSP.
Ref – MS: 3371-4800-1 (DMTE 2-8-2/RDIMS 184923).

Chapter 1 was not affected. Although PO 004 was to be removed, the submarine school decided to keep the same overall course timing. In order to do so, all the extra periods were diverted from PO 004 to PO 003. Sequence and Timings for PO 001 & 002 remained intact.

Chapter 2 was updated with the removal of PO 004. Chapter 3 was unchanged.
Chapter 4 had several changes, most notably with the removal of PO 004. All Timings were transferred evenly throughout PO 003. Some minor changes were also incorporated for all ECs and PCs to reflect proper timing for all trainee assessments.
Annex A was updated with current references. Those updates were in turn, transposed to all affected POs.

Annex B had T0004 removed from its list.

Annex D was updated with the removal of PO 004 from the Board Scalar.

Annex G was updated to reflect new timings for all POs, ECs and PCs for a course loaded with 6 students (optimum).

(April 2013) Version 1.4 incorporated changes to Chapter 1, removing CSE trades NWT and NET and replacing with new trades W Eng Tech 00366-04,05 and 07. Also, Sonar Op 00324 (AILE qualified MS and above) was added and MED Tech (PA) was changed to reflect all ranks.

(March 2022) Version 1.5 Reference to CFNOS changed to NFS(A)/NTDC(A)/NPTG HQ respectively. Physician Assistant updated to reflect new MOSID. Changes made to reflect fleet procedures, clarify teaching points, and reflect current adult learning practices. See CDAR for details.

CDAR CRWK(D) Mar2022.pdf

Note: This annex shall NOT be deleted. This annex must be annotated with every update / modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Detail as to the nature and reasons for amendments is to be provided.